

41227.

TEST - 2019.

Question 1

(a) $\text{mean} = \frac{\sum x_i}{n}$

ADVERTCH = -0.0240

PAN AFRICAN = 0.0212

- PAN AFRICAN HIGHER MEAN.

(b) $\text{STANDARD DEVIATION} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$

ADVERTCH = 0.0467

PAN AFRICAN = 0.0904

ADVERTCH $\rightarrow$ IT HAS A LOWER STANDARD DEVIATION.

(c) $CV = \frac{\sigma}{\mu}$

ADVERTCH = $\frac{0.0467}{-0.0240} = -1.9458$

PAN AFRICAN = $\frac{0.0904}{0.0212} = 4.2461$

ADVERTCH HAS A LOWER RISK PER UNIT OF RETURN $\rightarrow$ THEREFORE IT WOULD BE DESIRABLE.

$\Rightarrow$ C.V. USED IT SHOWS IS A MAJOR DIFFERENCE IN EXPECTED RETURNS.

(d). POSITIVE CORRELATION SUGGESTS THE TWO DO NOT PROVIDE ANY DIVERSIFICATION EFFECTS $\rightarrow$ THEREFORE AN INVESTOR SHOULD BE WORRIED ABOUT HAVING BOTH IN A PORTFOLIO.

Question two

a) $u(w) = \ln w$ $u'(w) = \frac{1}{w}$ ①

Absolute risk aversion ($A(w)$) = $-\frac{u''(w)}{u'(w)}$
= $\frac{1}{w}$ ①

$A'(w) = -\frac{1}{w^2}$ ①

Relative risk aversion ($R(w)$) = $-\frac{w u''(w)}{u'(w)}$ ①
= 1 ①

$R'(w) = 0$ ①

(b). The log utility function has a constant relative risk aversion of 1. ①
 $R'(w) = 0$.

⇒ Implies that investor keeps a constant proportion of wealth invested in risky assets as she gets richer. ①

Question THREE

(a). Absolute dominance exists, when one investment portfolio provides higher return than another in all possible circumstances. (1)

(b). FOSD theorem states that, assuming an investor prefers more to less, A will dominate B (i.e. investor will prefer portfolio A to B) if. (1)

$$F_A(x) \leq F_B(x) \text{ for all } x, \text{ and. (1/2)}$$

$$F_A(x) < F_B(x) \text{ for some value of } x. (1/2)$$

(c). SOSD theorem applies when the investor is risk averse, as well as preferring more to less. (1)

In this case A will dominate B if. (1)

$$\int_a^x F_A(y) dy \leq \int_a^x F_B(y) dy \quad \forall x. (1/2)$$

with the strict inequality holding for some value of x . (1/2)

and where, a , is the lowest return that the portfolios can possibly provide. (1/2)

(d) CDF.

	-5%	-3%	0	3%	5%
1	0.2	0.4	0.6	0.8	1
2	0.3	0.5	0.6	0.8	1

Scf.	-5%	-3%	0	3%	5%
1	0	-0.004	0.016	0.034	0.05
2	0	0.006	0.021	0.039	0.055

⇒ ASSET 1 HAS FOSD OVER ASSET 2.
 \$ STUDENT MAY USE THE BROADH USED IN CLASS.

QUESTION FOUR.

$$\begin{aligned} R_1 - R_f &= \delta_{11} z_1 + \delta_{12} z_2 \\ R_2 - R_f &= \delta_{21} z_1 + \delta_{22} z_2 \end{aligned}$$

$$\Rightarrow 0.03 = (0.2)^2 z_1$$

$$0.08 = (0.4)^2 z_2$$

$$\Rightarrow z_1 = 0.75$$

$$z_2 = 0.5$$

$$\Sigma z_i = 1.25$$

$$w_1 = \frac{0.75}{1.25} = 0.6$$

$$w_2 = 0.4$$

$$\begin{aligned} R_p &= 0.6(0.07) + 0.4(0.12) \\ &= 0.09 \text{ or } 9\% \end{aligned}$$

$$\begin{aligned} \sigma_p^2 &= (0.6)^2 (0.2)^2 + (0.4)^2 (0.4)^2 \\ &= 0.04 \end{aligned}$$

$$\sigma_p = 0.20 \text{ or } 20\%$$

b) WITH 40% ON TARRANTY AND 60% ON
 RISK FREE. ⇒ $E(r) = 6\%$ $\sigma = 8\%$
 ∴ ONE WOULD NEVER PREFER OLD MUTUAL FUND.

Question five

(a) From Asset φ . $R_s = R_f + \beta(R_m - R_f)$

$$18\% = 6\% + 1.5(R_m - 6\%)$$

$$R_m = \underline{14\%}$$

(b) Let market capitalization of φ be x .

$$\Rightarrow \frac{2.6}{13+x}(11\%) + \frac{39}{13+x}(10\%) + \frac{5.2}{13+x}(13\%) + \frac{x}{13+x}(18\%) + \frac{1.3}{13+x}(16\%) = 18\%$$

$$\therefore x = \underline{6.5m}$$

$$\beta = \frac{R_i - R_f}{R_m - R_f}$$

$$\beta_1 = \frac{8}{8}, \beta_2 = \frac{1}{2}, \beta_3 = \frac{7}{8}, \beta_5 = \frac{8}{4}$$

$0.1625 \quad 0.5 \quad 0.875 \quad 1.125$

(c) $\beta_p = \frac{19}{24}$ or 0.791667

- (d) ① Empirical studies do not provide strong support.
- ② Underlying assumptions not realistic.
- ③ Investors do not necessarily borrow and lend unlimited amounts at the same risk-free rate.
- ④ Markets for risky assets may not be perfect.
- ⑤ Investors may not have homogeneous expectations.
- ⑥ Does not account for taxes and transaction costs.

Question 8p

(a) b_{ik} - sensitivity of return of i to factor k

λ_2 = risk premium per unit of exposure corresponding to factor k .

λ_0 = risk-free rate.

(b) need to solve the linear system.

$$0.18 = 0.075 + 1.5\lambda_1 + 0.5\lambda_2$$

$$0.15 = 0.075 + 0.5\lambda_1 + 1.5\lambda_2$$

$$\Rightarrow \lambda_1 = 0.06 \quad \lambda_2 = 0.03$$

(c) Given portfolio returns as arbitrage opportunity, as the return does not satisfy the APT equation.

$$\text{Indeed } E(R_3) = 0.075 + 0.06 \times 0.75 + 0.03 \times 0.7 \\ = \underline{\underline{14.1\%}}$$

Therefore it is not a feasible portfolio. under assumption of no arbitrage.